

PUV Nearshore Wave Measurement Pipeline

Processing, Validation, and Spectral Bias Investigation

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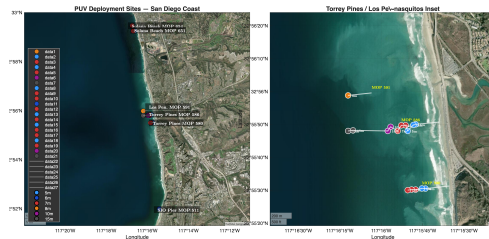
Outline

- 1 Overview
- 2 L1 Processing
- 3 L2 Spectral Analysis
- 4 Validation Against MOP
- 5 Hypothesis Testing
- 6 L3 Forcing Characterization
- 7 Key Findings and Next Steps

- Nearshore wave transformation governs beach morphology, sediment transport, and coastal hazards
- Accurate in-situ wave measurements at 5–15 m depth are needed to validate spectral wave models and drive transport calculations
- PUV (pressure–velocity) measurements from bottom-mounted ADVs provide directional spectra without surface moorings
- Goal: build a robust, reproducible processing pipeline from raw Nortek Vector data to validated wave statistics

Instrument Network

- **20 deployments, 40 instruments**
- 5 sites along San Diego coast:
 - Torrey Pines — MOP 580 & 586
 - SIO Pier — MOP 511
 - Solana Beach — MOP 654
 - Los Peñasquitos Lagoon — MOP 591
- Depths: 5–15 m
- Deployment period: 2023–2026
- Nortek Vector ADVs, 2 Hz continuous



Pipeline Architecture

① L1 — Raw to QC

Ingest burst files → clock drift correction → quality control → tilt correction → coordinate rotation

② L2 — Spectral Analysis

Segmentation → Welch PSD / cross-spectra → pressure correction → bulk parameters, bed velocity, stresses

③ Validation

PUV vs. CDIP MOP wave model, spectral bias investigation

④ Analysis scripts

Cross-deployment diagnostics, hypothesis testing

GitHub: [holdenlesliebole/Nortek_Vector_PUV_Pipeline](https://github.com/holdenlesliebole/Nortek_Vector_PUV_Pipeline)

Raw Data Ingestion

- Nortek Vector burst files: `.dat` (data), `.sen` (sensor/diagnostics), `.hdr` (header)
- Parsed with MATLAB `textscan` for speed — deployments can exceed 10^8 samples
- Continuous 2 Hz time series reconstructed from burst boundaries
- **Clock drift correction:** linear interpolation between deployment and recovery clocks
- Battery cutoff detection: $\text{gaps} > 1 \text{ s} \Rightarrow$ discard remainder of record

Quality Control — Old vs. New Approach

Old: Fixed threshold

- Absolute $\pm 5^\circ$ pitch/roll cap
- Discards all tilted data
- Problematic when pipes bend during storms
— stable but non-zero tilt is *correctable*

New: Variability-based QC

- Rolling std of pitch/roll over 60 s windows
- Flag samples where variability $> 2^\circ$
- Absolute cap retained at 30° for severely compromised instruments
- Stable tilts *retained* and corrected

Tilt Correction

Sample-by-sample 3D rotation using measured pitch (ϕ) and roll (θ):

$$\mathbf{R} = \mathbf{R}_{\text{roll}}(\theta) \mathbf{R}_{\text{pitch}}(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}$$

- Applied *before* heading rotation to buoy coordinates
- Corrects velocity components for instrument tilt
- Critical for instruments with bent mounting pipes (post-storm)

Coordinate Rotation

- Instrument XYZ $\rightarrow$ geographic buoy coordinates
- Convention: $+x$ West, $+y$ North, $+z$ Up (left-handed, MOP-consistent)
- Heading from deployment notes + magnetic declination via IGRF-13 model at deployment location and epoch
- Shore-normal rotation applied later in L2 using CDIP THREDDS shore-normal angle for each MOP station

33 / 40 instruments processed successfully (**82.5%**)

7 failures — all hardware-related:

- Instrument burial (sand accretion)
- Bent or broken mounting pipes
- Dead / depleted batteries (no usable data)

No failures attributable to the processing pipeline itself.

- **Segment length:** 2048 samples = 17.1 min at 2 Hz
- Non-overlapping segments
- **NaN policy:** segments with $>10\%$ NaN discarded; $\leq 10\%$ filled by linear interpolation
- Segment-mean pressure $\rightarrow$ water depth H
- All channels linearly detrended before spectral analysis

Spectral Estimation

- **Method:** Welch's averaged periodogram
- Sub-segments: $N_{\text{fft}} = 256$ samples (128 s)
- **Window:** Hanning, 50% overlap
- ≈ 34 **effective DOF** per segment
- Frequency resolution: $\Delta f \approx 0.0078$ Hz, range 0–1 Hz
- **Cross-spectra:** S_{pu} , S_{pv} computed with same windowing for directional coefficients a_1 , b_1

Pressure Correction

Surface elevation spectrum from pressure:

$$S_{\eta}(f) = \frac{S_p(f)}{K_p^2(f)}, \quad K_p(f) = \frac{\cosh(k z_s)}{\cosh(k H)}$$

- Wavenumber $k(f)$: linear dispersion relation $\omega^2 = gk \tanh(kH)$ solved via Newton's method
- **Cutoff**: $K_p < 0.1 \Rightarrow$ set $S_{\eta}(f) = 0$ (noise suppression)
- Resulting f_{cut} recorded per segment
- Verified: Newton, Wu (1986), and O'Reilly formulations produce identical K_p (see Validation section)

Bulk Wave Parameters

- $H_s = 4\sqrt{m_0}$, computed separately for:
 - Sea-swell: 0.04–0.25 Hz
 - Infragravity: 0.004–0.04 Hz
- Peak period $T_p = 1/f_p$ (max S_η in SS band)
- Mean period $T_{m02} = \sqrt{m_0/m_2}$
- Mean direction from a_1 , b_1 (PUV cross-spectra)
- Energy flux $F = \rho g \int c_g(f) S_\eta(f) df$

Additional L2 Products

- **Near-bed velocity:** IFFT method — scale velocity FFT by $\cosh(kH)/\cosh[k(H - z_s)]$, inverse transform
- **Bed shear stress:** Swart (1974) friction factor, $k_s = 2.5 D_{50}$
- **Reynolds stress:** $\overline{u'w'}$, $\overline{v'w'}$, $\overline{u'v'}$; turbulent KE
- **Velocity moments:** skewness S_k , asymmetry A_s (Elgar 1985; Hoefel & Elgar 2003)

Shore-Normal Rotation

- Shore-normal angle retrieved from CDIP THREDDDS for each MOP station
- Velocities rotated: $+x$ onshore, $+y$ alongshore north
- **Fallback:** if THREDDDS unavailable, retain geographic buoy coordinates ($+x$ W, $+y$ N)
- Shore-normal frame required for cross-shore energy flux and transport calculations

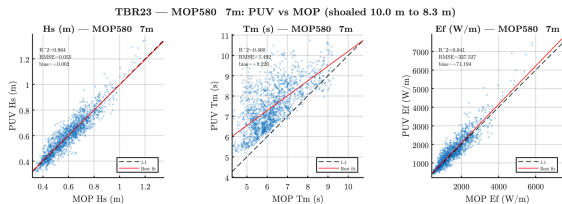
- CDIP Monitoring and Prediction (MOP) spectral wave model
- Initialized by offshore CDIP buoy observations
- Provides $S_{\eta}(f)$ at 10 m depth, hourly
- **Shoaling to PUV depth:**

$$S_{\eta}(f, h_{\text{PUV}}) = S_{\eta}(f, 10 \text{ m}) \times \frac{c_g(f, 10 \text{ m})}{c_g(f, h_{\text{PUV}})}$$

- PUV segments (17 min) interpolated to MOP hourly time base for comparison

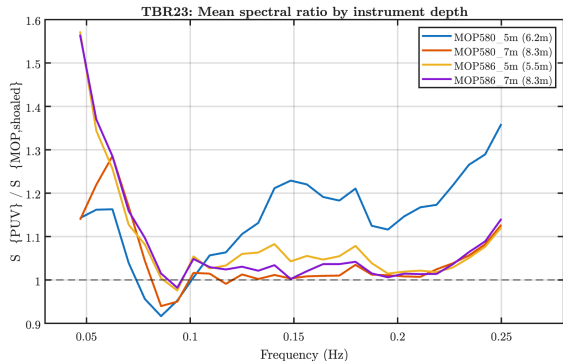
Bulk Comparison Results

- H_s : $R^2 = 0.83\text{--}0.86$
- $\text{RMSE} < 0.06\text{ m}$
- Small positive bias ($\text{PUV} > \text{MOP}$)
- Kelp-fouled instrument: $R^2 = 0.66$, bias $= +0.037\text{ m}$



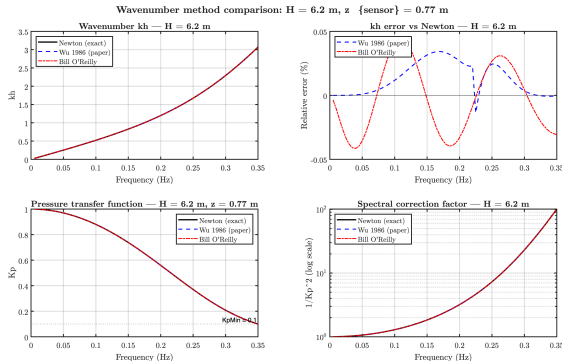
Frequency-Dependent Bias

- Excess energy concentrated at **swell peak** (0.04–0.08 Hz)
- $K_p > 0.9$ at these frequencies — pressure correction is minimal
- Mid-swell band: near-unity ratio (0.99–1.03)
- **NOT** a high-frequency pressure correction artifact



Pressure Correction Investigation

- Compared three wavenumber formulations:
 - 1 Newton's method (baseline)
 - 2 Wu (1986) approximation
 - 3 O'Reilly coefficients
- All produce **identical** K_p
- Pressure correction is **not** the source of the observed bias



Overview: Four Hypotheses

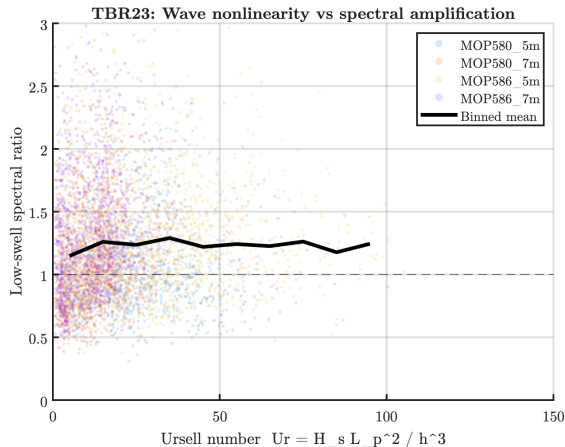
What causes the systematic positive H_s bias ($PUV > MOP$)?

- ① **Nonlinear shoaling** — triad interactions (Ursell number)
- ② **Directional narrowing** — refraction-driven focusing
- ③ **Bound long waves** — group-forced IG contamination
- ④ **MOP spectral peak broadening** — numerical diffusion in the wave model

H1: Nonlinear Shoaling

- Test: correlate spectral ratio with Ursell number $Ur = H_s L_p^2 / h^3$
- $N = 5228$ segment-hours
- **Result:** $R^2 = 0.004$
- Median Ur : 12 at 8.3 m, 30 at 5.5 m

⇒ RULED OUT



H2: Directional Narrowing

- Refraction should *narrow* directional spread during shoaling $\Rightarrow$ energy focusing $\Rightarrow$ higher H_s
- **Result:** PUV spread is systematically *wider* than MOP at swell frequencies (15° vs. 10°)
- Opposite to prediction
- Correlation with H_s amplification: $R^2 < 0.001$

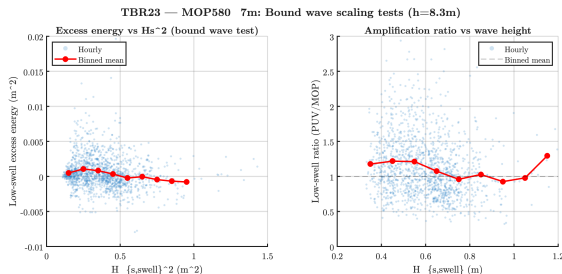
$\Rightarrow$ RULED OUT

Wider PUV spread may reflect scattered energy at shallow depth not captured by the forward-propagating MOP model.

H3: Bound Long Waves

- Bound wave energy $\propto H_s^2$ (Longuet-Higgins & Stewart 1962)
- **Result:** no positive correlation with H_s^2 ($R = -0.14$)
- Excess in *wrong frequency band* — at swell peak (0.04–0.08 Hz), not at group frequencies (0.01–0.03 Hz)

⇒ RULED OUT

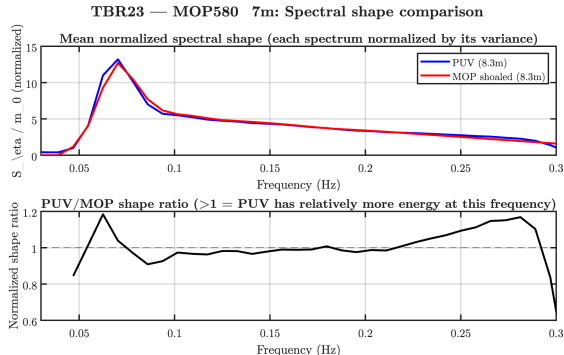


H4: MOP Spectral Peak Broadening

- Q_p (Goda peakedness): PUV 7% higher than MOP
- Half-power bandwidth: PUV **16.7% narrower**
- Peak spectral density: PUV 14.4% higher
- ΔQ_p vs. H_s amplification: $R = 0.44$, $R^2 = 0.20$

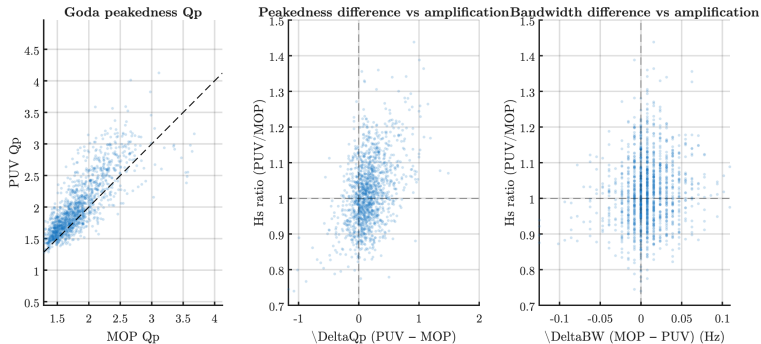
⇒ **SUPPORTED**

MOP model introduces numerical broadening of the spectral peak during propagation; PUV captures the true, narrower peak shape.



Peakedness Diagnostic

TBR23 — MOP580 7m: Spectral peakedness analysis



Hours when PUV spectrum is more peaked relative to MOP correspond to hours with larger positive H_s bias.

Hypothesis Summary

Hypothesis	Test metric	Result	Verdict
1. Nonlinear shoaling	$R^2(Ur, \text{ratio})$	$R^2 = 0.004$	Ruled out
2. Directional narrowing	PUV vs. MOP spread	PUV wider	Ruled out
3. Bound long waves	$R(H_s^2, \text{excess})$	$R = -0.14$	Ruled out
4. Peak broadening	$R(\Delta Q_p, \text{bias})$	$R = 0.44$	Supported

Cross-Deployment Confirmation

- Pattern confirmed across **11 instruments** from multiple deployments
- Correlation $R(Q_p) = 0.30\text{--}0.63$ across instruments
- Spans:
 - Summer and winter wave climates
 - Depths 5.5–15.5 m
 - Multiple MOP transects
- Spectral peak broadening is a **robust, general** feature of the MOP model — not site- or season-specific

L3 Architecture

- **L3a:** Frequency-band energy decomposition (IG / swell / sea)
- **L3b:** Storm event detection with MOP gap-filling
- **L3c:** Transport proxies (F_b , Shields, Rouse, mobilization)
- **L3d:** Tidal current decomposition (t_tide + NOAA depth)
- All components verified: $E_{f_total}/L2.E_f = 1.0000$ for all 33 instruments

Seasonal Energy Partitioning

- **Winter:** swell-dominated (65–90% of energy flux)
- **Summer:** sea-dominated or equal ($\sim 50/50$)
- **Spring:** transitional
- Consistent across all 5 sites

Storm Detection + MOP Gap-Filling

- NN24 Dec 28, 2023: H_s up to 3.6 m, 74-hour duration
- MOP fills data gaps from instrument damage during storms
- Example: MOP651_7m detected 0 PUV storms, MOP found 3 (86-hour event)
- Critical for characterizing the forcing that damaged the instruments

Transport Proxy Depth Dependence

- Shields parameter decreases offshore: 0.24 (5.6 m) to 0.065 (15.5 m)
- Mobilization: 100% at 5 m, 78% at 15 m (winter)
- Rouse number: bedload-dominated across all sites
- Cumulative F_b : winter $3\times$ summer at equivalent depth

Tidal Validation

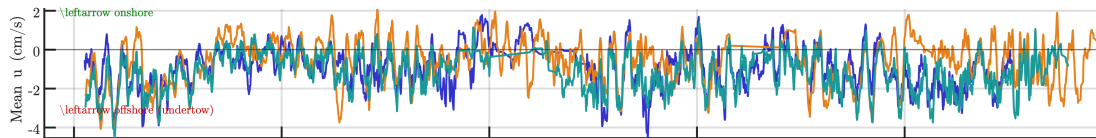
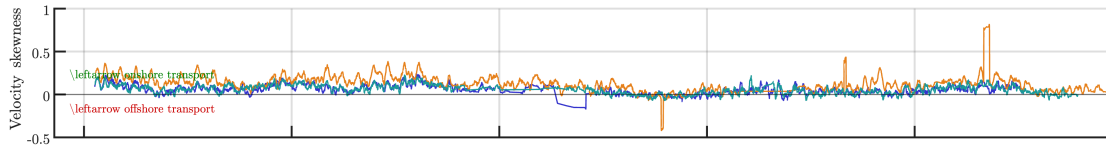
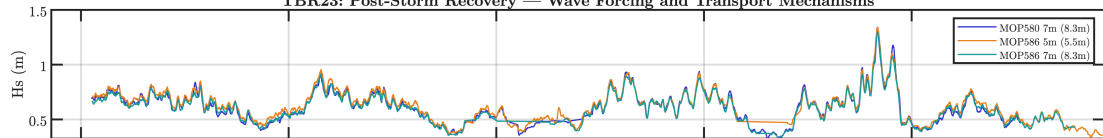
- NOAA Scripps Pier tide gauge: $R = 0.995$ with PUV depth
- Confirms PUV clocks are UTC (lag = 0.0 hours)
- Subtidal cross-shore current: consistently offshore (undertow)
- Tidal currents small (<0.05 m/s) vs. wave-driven flows

TBR23 Context: Post-Storm Recovery

- Winter 2023 storms deposited ~ 85 cm sand at 5 m depth
- TBR23 deployment (May–Aug 2023) captures recovery phase
- 22 jetski surveys + 4 PUV instruments at 5 m and 7 m depth
- **Question:** what wave processes drive the recovery?

Wave Forcing and Bed Response

TBR23: Post-Storm Recovery — Wave Forcing and Transport Mechanisms

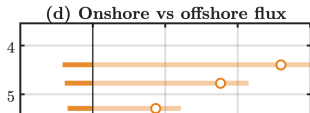
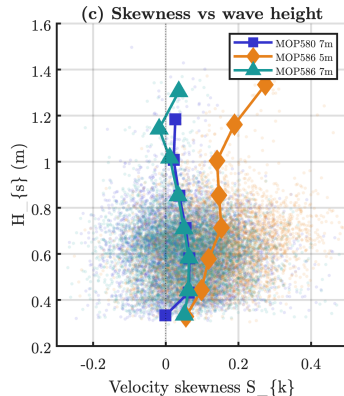
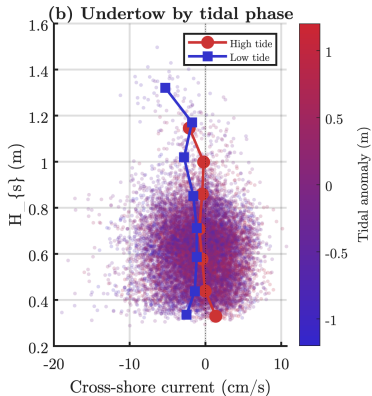
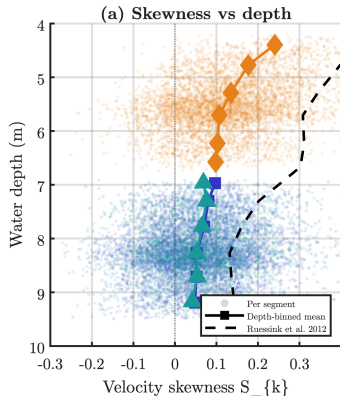


Competing Transport Mechanisms

- **Onshore: velocity skewness** ($S_k > 0$) — peaked crests drive bedload shoreward
- **Offshore: undertow** (mean return flow) — opposes recovery
- Both present simultaneously at all instruments
- Balance determines net transport direction

Transport Analysis

TBR23: Competing Transport During Post-Storm Recovery



Depth range from tidal modulation:

Each PUV is at a fixed cross-shore position.
The ~2.5m tidal range modulates the water

Recovery Boundary at 6–7 m Depth

- At 5.5 m: skewness flux 3–4 $\times$ stronger than undertow $\rightarrow$ net onshore (recovery)
- At 8.3 m: undertow slightly dominates $\rightarrow$ near-equilibrium
- Crossover at $\sim$ 6–7 m depth
- Explains: mid-July recovery pulse at 5 m site but not at 7 m sites
- Tidal modulation provides depth range at each fixed instrument

Key Physics

- Skewness follows Ruessink et al. (2012) depth parameterization
- Undertow strengthens at high tide (more wave energy at depth)
- Low tide reduces undertow, enabling onshore recovery
- MOP580_5m excluded: kelp fouling produced artifact negative S_k

Key Findings

- ① **Pipeline validated:** 33/40 instruments processed (82.5%), strong agreement with MOP ($R^2 > 0.83$, $RMSE < 0.06\text{ m}$)
- ② **Bias identified:** small systematic positive H_s bias concentrated at the swell spectral peak, *not* from the pressure correction
- ③ **Mechanism determined:** MOP model introduces numerical spectral peak broadening; PUV captures the true narrower peak ($R = 0.44$ between ΔQ_p and bias)
- ④ **Cross-deployment robustness:** pattern confirmed across 11 instruments, summer/winter, 5.5–15.5 m depth
- ⑤ **Full L1–L3 pipeline operational:** all 33 instruments processed through forcing characterization, storm detection, and transport proxy computation
- ⑥ **Full TBR23 analysis:** identified competing skewness vs. undertow transport, recovery boundary at 6–7 m depth

Implications for TBR23 Paper

- Pipeline is validated and ready for production use
- Bias is understood and attributable to the model, not the measurement — no correction needed for PUV data
- Spectral shape finding *supports* using PUV as ground truth for nearshore wave characterization
- L2 products (bed velocity, stress, moments) are ready for sediment transport analysis

Next Steps

- **Bailard / Hoefel & Elgar transport model:** implementation for quantitative flux estimates
- **L5 PUV–altimeter integration:** reconciled plan ready for deployment
- **IG / bound-wave analysis (L4):** separate forced vs. free IG energy
- **TBR23 paper draft:** synthesize validation, transport, and morphology results

Questions?